

Lemma [Core] 6.1: $\mathfrak{F} \text{ Core} \Rightarrow (\text{M-S EP}) \text{ for } \mathcal{I}_{\mathfrak{F}}$

If $\mathfrak{F}$ is a Core FD Complex, then if $f \in \phi_{\mathfrak{F}}(K \cup e) - \phi_{\mathfrak{F}}(K)$ then $e \in \phi_{\mathfrak{F}}(K \cup f) - \phi_{\mathfrak{F}}(K)$ for all $e, f \in E_{\mathfrak{F}}$ and $K \in \mathcal{I}_{\mathfrak{F}}$.

Proof: Suppose $f \in \phi_F(K \cup e) - \phi_F(K)$. If $e = f$ then $e \in \phi_F(K \cup f) - \phi_F(K)$.

For $e \neq f$, neither e nor f is in $\phi_{\mathfrak{F}}(K)$, because $\phi_{\mathfrak{F}}(K \cup e) - \phi_{\mathfrak{F}}(K) \neq \emptyset$. By **Lemma [KyCI] 4.5 (c)**, neither $K \rightarrow e$ nor $K \rightarrow f$ is in $F_{\mathfrak{F}}$. We have: $K \in \mathcal{I}_{\mathfrak{F}}$ and $e, f \in E_{\mathfrak{F}} - K$.

Hence both $K \cup e$ and $K \cup f$ are in $\mathcal{I}_{\mathfrak{F}}$ by strong extension (FD 4 contra.). Moreover, $K \cup e$ is a key of $\phi_{\mathfrak{F}}(K \cup e)$; thus $K \cup e$ is a maximal independent set in $\phi_{\mathfrak{F}}(K \cup e)$ by **Lemma [KyCI] 4.5**. This means $(K \cup e) \cup f$ is dependent.

Let C be an md-set contained in $(K \cup e) \cup f$. Both e and f are in C because $K \cup e$ and $K \cup f$ are independent. $C - e \subseteq \phi_F(K \cup f)$. This means, $e \in \phi_F(K \cup f)$ by **Theorem [Core] 6.5 Theorem (5)**. $\therefore e \in \phi_F(K \cup f) - \phi_F(K)$ because $e \notin \phi_F(K)$.

□

Theorem [Core] 6.2: $\text{Core} \equiv \text{Matroid}$

$\mathfrak{F}$ is a Core FD Complex if and only if $\phi_{\mathfrak{F}}$ satisfies the Matroid Closure Axioms.

Proof: Since $\phi_{\mathfrak{F}}$ satisfy the axioms for closure operators, we need only show $\phi_{\mathfrak{F}}$ satisfies strong extension whenever $\phi_{\mathfrak{F}}$ satisfies (M-S EP).

(Strong Extension $\Rightarrow$ (M-S EP))

If a Core Complex $\mathfrak{F}$ satisfies (M-S EP) on its independent sets then $\mathfrak{F}$ satisfies (M-S EP),

Lemma [Core] 6.9. $\mathfrak{F}$ is a Core Complex implies, $\mathfrak{F}$ satisfies (M-S EP) on independent sets, **Lemma [Core] 6.10**. Thus core FD-Complexes meet (M-S EP).

((M-S EP) $\Rightarrow$ Strong Extension)

This says that every Core FD Complex satisfies (M-S EP).

((M-S EP) $\Rightarrow$ Strong Extension)

Suppose $e \notin \phi_{\mathfrak{F}}(K)$ for a $K \in \mathcal{I}_{\mathfrak{F}}$ and $K \cup e \notin \mathcal{I}_{\mathfrak{F}}$. Weak extension gives a key $K' \subseteq K \cup e$ such that K' is a key of $\phi_{\mathfrak{F}}(K \cup e)$ and $e \in K'$. K' is a proper subset of $K \cup e$ because $K \cup e$ is dependent. There is an $f \in K \cup e$ such that $f \notin K'$ and $K' \rightarrow f \in \mathfrak{F}$. $f \neq e$ because $f \notin K'$. Now, $f \in \phi_{\mathfrak{F}}(K')$ because $f \in K \cup e \subseteq \phi_{\mathfrak{F}}(K')$ and $f \notin \phi_{\mathfrak{F}}(K' - e)$ because $f \subseteq K$, $f \notin K'$ and $\phi_{\mathfrak{F}}(K' - e) \cap K = K' - e \subseteq K'$. Note that if the last intersection includes